



# 8255 Programmable Peripheral Interface

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## ➔ 6.1 8255 | Introduction

1. 8255 is a Programmable Peripheral Interface.
2. It is used to interface (connect)  $\mu$ P with I/O devices such as a printer, keyboard etc. 8255 has three 8 bit bidirectional I/O ports called Port A, Port B and Port C.
3. They can be used as input ports or as output ports.
4. These ports can operate in three different modes of data transfer.
5. Mode 0 is called Simple I/O. Here the three ports perform basic 8 bit data transfers.
6. Mode 1 is called Handshake I/O. Here port A and Port B transfer data using an advanced technique called handshaking. This prevents any loss of data and hence makes the transfer more reliable. To carry out handshaking, lines of Port C are used up by Port A and Port B.
7. Mode 2 is bidirectional handshaking in which as the name suggests, the port can perform input as well as output handshaking.
8. The modes are selected by the programmer by giving commands to 8255. Depending upon the system requirement, the appropriate command is formed and is first stored in a register like AL. Then using an instruction like OUT, the command is sent to 8255 through the data bus.
9. Additionally, 8255 has an excellent feature called BSR command. It stands for Bit Set Reset. Using BSR command, the programmer can individually set or reset (send 1 or 0) to any line of Port C without affecting other lines. This basically transforms Port C from being one 8 bit port to becoming eight 1 bit ports that can be individually controlled by the programmer. This feature is very useful in complex interfaces like ADC, LCD controller etc.

## ➔ 6.2 8255 | Sallent Features

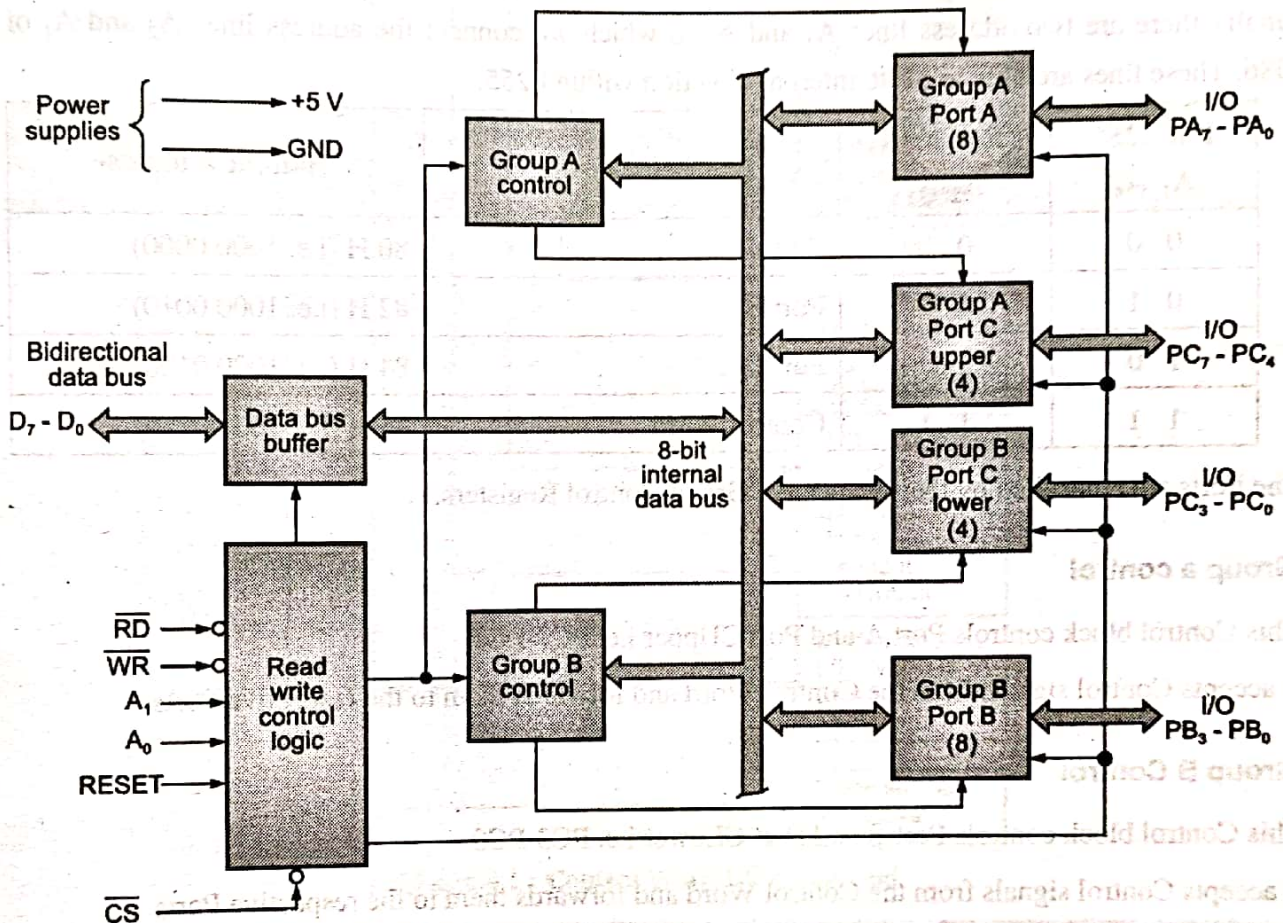
1. It is a **programmable** general-purpose I/O device.
2. It has 3 8-bit bi-directional I/O ports: Port A, Port B, and Port C.
3. It provides 3 modes of data transfer: Simple I/O, Handshake I/O and Bi-directional Handshake.
4. Additionally it also provides a Bit Set Reset Modes to alter individual bits of Port C.



## ✓ Syllabus Topic : 8255-PPI – Block Diagram

## 6.3 8255 | Architecture | Internal Block Diagram

- Q. Explain with block diagram working of 8255 PPI. (Dec. 15, 10 Marks)
- Q. Explain PPI 8255 with block diagram. (May 18, 10 Marks)
- Q. Draw and explain the block diagram of 8255 Programmable Peripheral Interface (PPI) with control word formats. (May 19, 10 Marks)



(1c) Fig. 6.3.1 : Architecture of 8255

The architecture of 8255 has the following main components...

#### Data bus buffer

- This is a 8-bit bi-directional buffer used to interface the internal data bus of 8255 with the external (system) data bus.
- The CPU transfers data to and from the 8255 through the data bus via this buffer.
- Moreover, the commands given to 8255 (I/O command and BSR command) are also given through the same data bus.



**Read/Write control logic**

- It accepts address and control signals from the  $\mu P$ .
- The Control signals determine whether it is a read or a write operation.
- During a read, data will be transferred from 8255 to the  $\mu P$ .
- During a write, data will be transferred from  $\mu P$  to 8255.
- There is a chip selection signal that selects 8255 on the basis of its address.
- The reset signal is to reset 8255 and stop all current transfers.
- Finally there are two address lines  $A_1$  and  $A_0$  to which we connect the address lines  $A_2$  and  $A_1$  of the  $\mu P$  8086. These lines are used to make internal selection within 8255.

| For 8255<br>$A_1 A_0$ | For 8086<br>$A_2 A_1$ | Selection    | Sample address        |
|-----------------------|-----------------------|--------------|-----------------------|
| 0 0                   | 0 0                   | Port A       | 80 H (i.e. 1000 0000) |
| 0 1                   | 0 1                   | Port B       | 82 H (i.e. 1000 0010) |
| 1 0                   | 1 0                   | Port C       | 84 H (i.e. 1000 0100) |
| 1 1                   | 1 1                   | Control Word | 86 H (i.e. 1000 0110) |

The Ports are controlled by their respective Group Control Registers.

**Group A control**

- This Control block controls Port A and Port C Upper i.e. PC7-PC4.
- It accepts Control signals from the Control Word and forwards them to the respective Ports.

**Group B Control**

- This Control block controls Port B and Port C Lower i.e. PC3-PC0.
- It accepts Control signals from the Control Word and forwards them to the respective Ports.

**Port A, Port B, Port C**

- These are 8-bit Bi-directional Ports.
- They can be programmed to work in the various modes as follows :

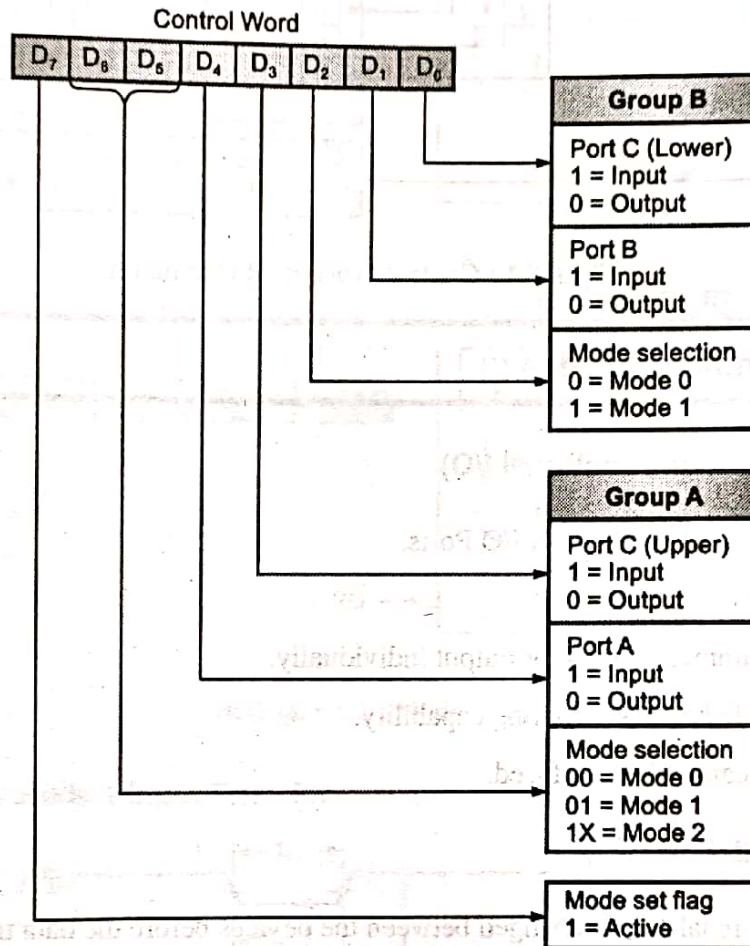
| Port   | Mode 0 | Mode 1                 | Mode 2                 |
|--------|--------|------------------------|------------------------|
| Port A | Yes    | Yes                    | Yes                    |
| Port B | Yes    | Yes                    | No (Mode 0 or Mode 1)  |
| Port C | Yes    | No (Handshake signals) | No (Handshake signals) |

- ONLY Port C can also be programmed to work in Bit Set reset Mode to manipulate its individual bits.

## 6.4 Control Word of 8255 - I/O Mode (I/O Command)

Q. Explain the I/O mode control word format of 8255 PPI. (Dec. 18, 5 marks)

- To do 8-bit data transfer using the Ports A, B or C, 8255 needs to be in the IO mode.
- The bit pattern for the control word in the IO mode is as follows :



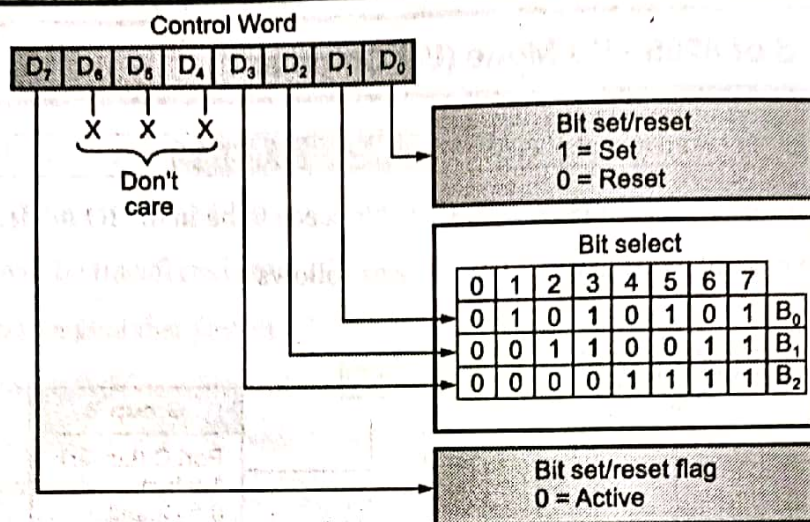
(1c2)Fig. 6.4.1 : Control Word I/O Command

## 6.5 Control Word of 8255 - BSR Mode (BSR Command) {ONLY for Port C}

Q. Discuss control word format for Bit Set Reset (BSR) mode of 8255 PPI. (Dec. 17, 5 Marks)

- The BSR Mode is used **ONLY** for Port C.
- In this Mode the **individual bits** of Port C can be set or reset.
- This is very useful as it provides **8 individually controllable lines** which can be used while interfacing with devices like an **A to D Converter** or a **7-segment display** etc.
- The individual bit is **selected** and Set/reset through the **control word**.
- Since the D7 bit of the Control Word is 0, the BSR operation will **not affect the I/O operations** of 8255.





(1c3) Fig. 6.5.1 : Control Word BSR Command

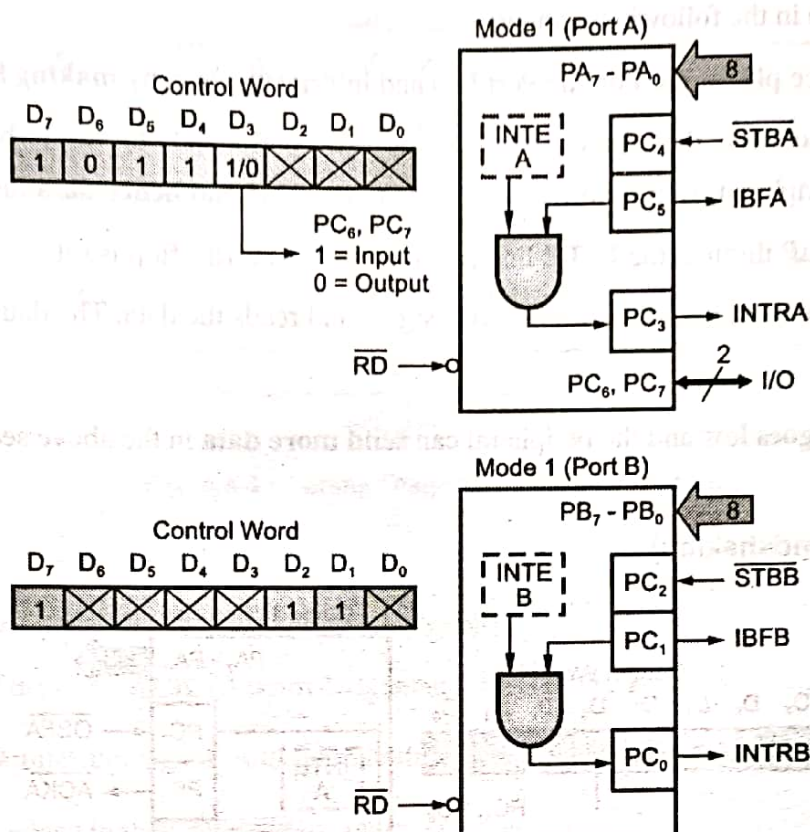
## 6.6 8255 | Data Transfer Modes

### 6.6.1 Mode 0 (Simple BI-directional I/O)

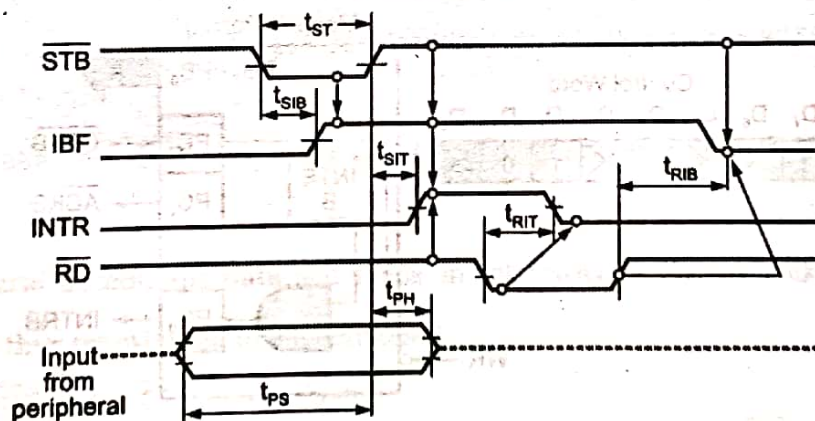
- Port A and Port B used as 2 Simple 8-bit I/O Ports.
- Port C is used as 2 simple 4-bit I/O Ports.
- Each port can be programmed as input or output individually.
- Ports do not have handshake or interrupting capability.
- Hence, slower devices cannot be interfaced.

### 6.6.2 Mode 1 (Handshake I/O)

- In Mode 1, handshake signals are exchanged between the devices before the data transfer takes place.
- Port A and Port B used as 2 8-bit I/O Ports that can be programmed in Input OR in output mode.
- Each Port uses 3 lines from Port C for handshake. The remaining lines of Port C can be used for simple IO.
- Interrupt driven data transfer and Status driven data transfer possible.
- Hence, slower devices can be interfaced.
- The handshake signals are different for input and output modes.

**Mode 1 (Input Handshaking)**

(1c4) Fig. 6.6.1 : Mode 1 Input

**Timing Diagram for Mode 1 Input Transfer**

(1c5) Fig. 6.6.2 : Timing Diagram for Mode 1 Input

**Working**

- Each port uses 3 lines of Port C for the following signals:
- STB (Strobe), IBF (Input Buffer Full) → Handshake signals
- INTR (interrupt) → Interrupt signal



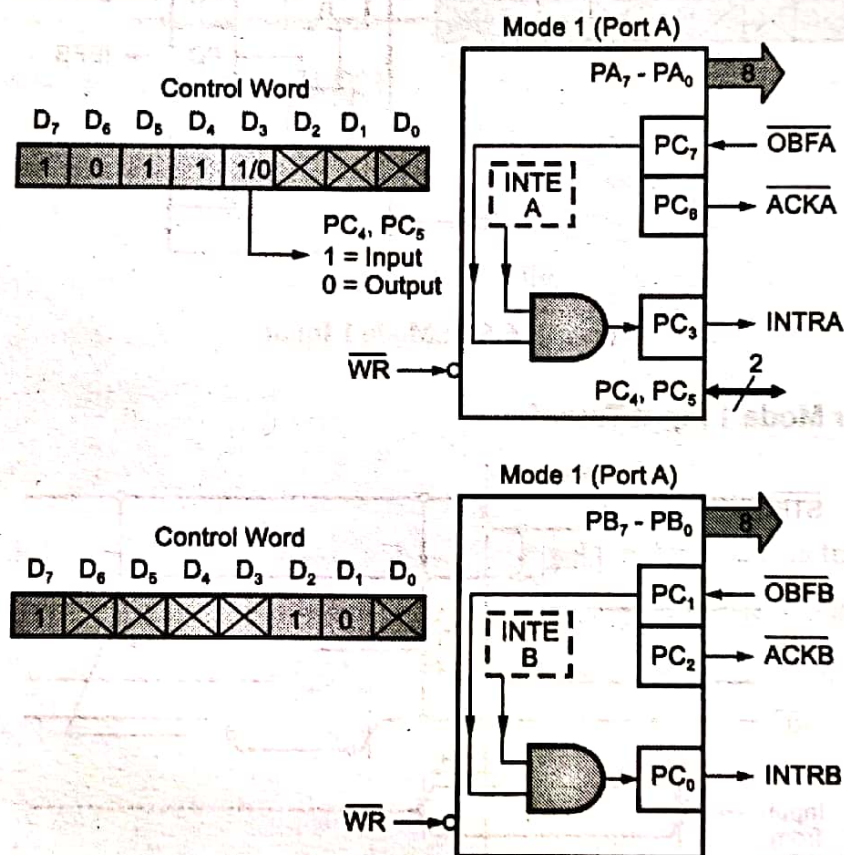


- Additionally the  $\overline{RD}$  signal of 8255 is also used.

**Handshaking** takes place in the following manner:

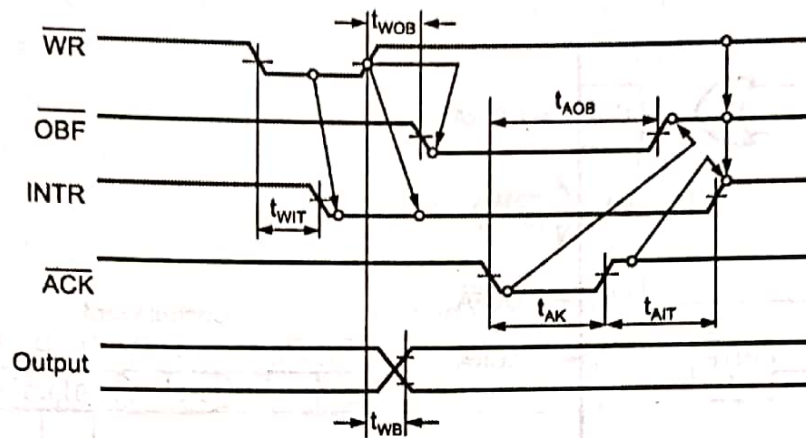
- (1) The peripheral device places data on the Port bus and informs the Port by making STB low.
- (2) The input Port accepts the data and informs the peripheral to wait by making IBF high. This prevents the peripheral from sending more data to the 8255 and hence data loss is prevented.
- (3) 8255 interrupts the  $\mu P$  through the INTR line provided the INTE flip-flop is set.
- (4) In response to the Interrupt, the  $\mu P$  issues the  $\overline{RD}$  signal and reads the data. The data byte is thus transferred to the  $\mu P$ .
- (5) Now, the IBF signal goes low and the peripheral can send more data in the above sequence.

### Mode 1 (Output Handshaking)



(1c) Fig. 6.6.3 : Mode 1 Output

## Timing Diagram for Mode 1 Output Transfer



(1C7)Fig. 6.6.4 : Timing Diagram for Mode 1 Output

### Working

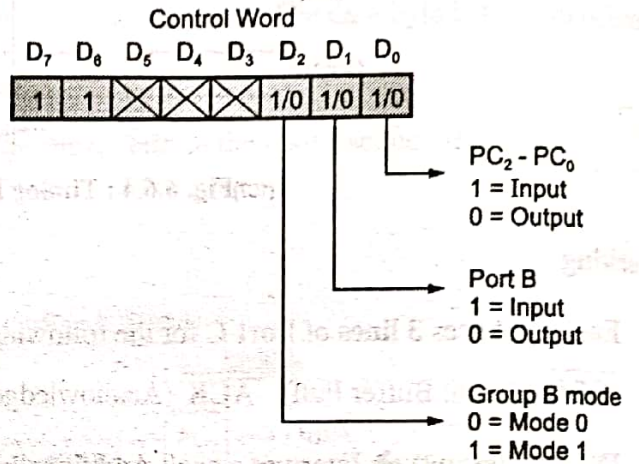
- Each port uses 3 lines of Port C for the following signals:
  - $\overline{\text{OBF}}$  (Output Buffer Full),  $\overline{\text{ACK}}$  (Acknowledgement)  $\rightarrow$  Handshake signals
  - $\text{INTR}$  (interrupt)  $\rightarrow$  Interrupt signal. Additionally the  $\overline{\text{WR}}$  signal of 8255 is also used.
  - Handshaking takes place in the following manner:
    - (1) When the output port is empty (indicated by a high on the  $\text{INTR}$  line), the  $\mu\text{P}$  writes data on the output port by giving the  $\overline{\text{WR}}$  signal.
    - (2) As soon as the  $\overline{\text{WR}}$  operation is complete, the 8255 makes the  $\text{INTR}$  low, indicating that the  $\mu\text{P}$  should wait. This prevents the  $\mu\text{P}$  from sending more data to the 8255 and hence data loss is prevented.
    - (3) 8255 also makes the  $\overline{\text{OBF}}$  low to indicate to the output peripheral that data is available on the data bus.
    - (4) The peripheral accepts the data and sends an acknowledgement by making the  $\overline{\text{ACK}}$  low. The data byte is thus transferred to the peripheral.
    - (5) Now, the  $\overline{\text{OBF}}$  and  $\overline{\text{ACK}}$  lines go high.
    - (6) The  $\text{INTR}$  line becomes high to inform the  $\mu\text{P}$  that another byte can be sent. i.e. the output port is empty.
- This process is repeated for further bytes.



### 6.6.3 Mode 2 (BI-directional Handshake I/O)

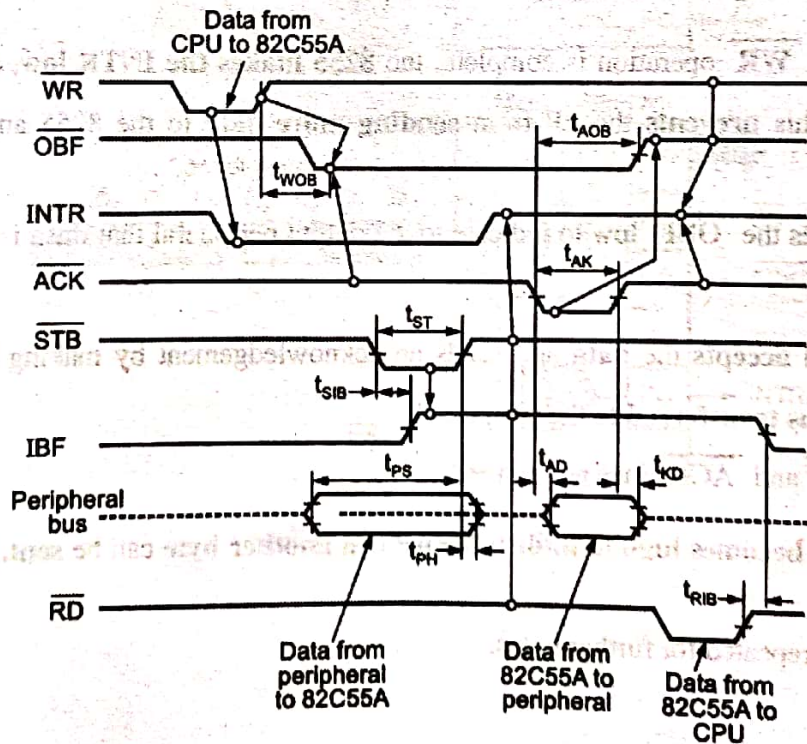


**(1c8)Fig. 6.6.5 : Mode 2**



**(1c9)Fig. 6.6.6 : Control Word Mode 2**

### 👉 Timing Diagram for Mode 2 Bi-Directional Transfer



**(1c10)Fig. 6.6.7 : Timing Diagram for Mode 2**



**Working:**

- In this mode, Port A is used as an 8-bit bi-directional Handshake I/O Port.
- Port A requires 5 signals from Port C for doing Bi-directional handshake.
- Port B has the following two options:
  - (1) Use the remaining 3 lines of Port C for handshaking so that Port B is in Mode 1. Here Port C lines will be completely used for handshaking (5 by Port A and 3 by Port B).
  - (2) Port B works in Mode 0 as simple I/O.

In this case the remaining 3 lines of Port C can be used for data transfer.

Port A can be used for data transfer between two computers as shown.

The high-speed computer is known as the master and the dedicated computer is known as the slave.

Handshaking process is similar to Mode 1.

**For Input:**

STB and IBF → handshaking signals, INTR → Interrupt signal.

**For Output:**

OBF and ACK → handshaking signals, INTR → Interrupt signal.

Thus the 5 signals used from Port C are:

STB, IBF, INTR, OBF and ACK.

✓ Syllabus Topic : Interfacing with 8086

## ➔ 6.7 8255 | Interfacing with 8086

- (1) 8255 is a programmable peripheral interface.

It is used to interface microprocessor with I/O devices via three ports: PA, PB, PC.

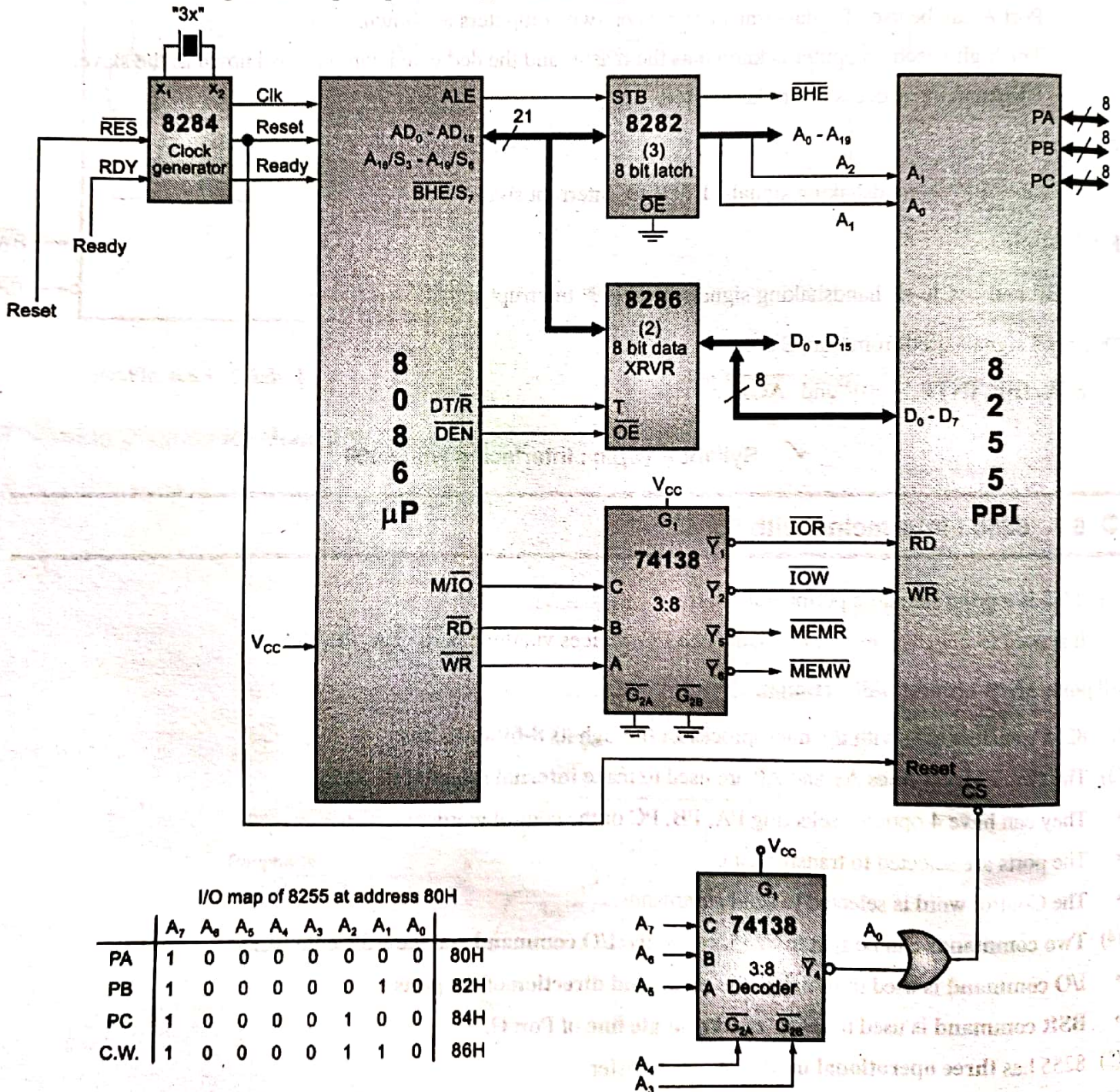
**All ports are 8-bit and bidirectional.**

- (2) 8255 transfers data with the microprocessor through its **8-bit data bus**.
- (3) The **two address lines** A1 and A0 are used to make **internal selection** in 8255.
  - They can have 4 options, selecting PA, PB, PC or the control word.
  - The ports are selected to transfer data.
  - The Control word is selected to send commands.
- (4) **Two commands** can be sent to 8255, called the **I/O command** and the **BSR command**.
  - **I/O command** is used to initialize the **mode and direction** of the ports.
  - **BSR command** is used to **set or reset** a single line of Port C.
- (5) 8255 has **three operational modes** of data transfer.
- (6) Mode 0 is a simple data transfer mode.





- It does not perform handshaking but all three ports are available for data transfer.
- (7) Mode 1 performs unidirectional handshaking.
  - That makes transfers more reliable.
  - Port C lines are used by Port A and Port B to perform Handshaking.
- (8) Mode 2 performs bidirectional handshaking.
  - Only Port A can operate in Mode 2.
  - At that time Port B can operate in Mode 1 or Mode 0.
  - Port C lines are again used up for performing Handshaking for Port A and Port B.



(11C11) Fig. 6.7.1 : 8255 Interfacing with 8086





## 6.8 8255 I Programming

### 6.8.1 Program to Initialize 8255

Q. WAP to initialize 8255 as follows :

Port A --- Mode1 ---- i/p, Port B --- Mode1 ---- o/p, Port C --- Handshaking Assume 8255 is at 80H.

Soln. :

```
Code  SEGMENT
      ASSUME CS: Code
      MOV AL, B4H
      OUT 86H, AL          ; Control Word = 1011 0100 = B4H
      INT 03H
Code  ENDS
      END
```

### 6.8.2 Program to Generate Square Wave using 8255

Q. Generate a square wave on a display device connected to PortC<sub>3</sub> by BSR command.

Soln. :

```
Code  SEGMENT
      ASSUME CS: Code
Back:  MOV AL, 06H
      OUT 86H, AL          ; BSR Command (Send 0) = 0000 0110
      CALL Delay
      MOV AL, 07H
      OUT 86H, AL          ; BSR Command (Send 1) = 0000 0111
      CALL Delay
      JMP Back
      INT 03H
Code  ENDS
      END
```





### 6.8.3 Program to Generate Positive Spikes

**Q.** WAP to generate "positive spikes" on a display device connected to PortC<sub>3</sub> using BSR command.

**Soln. :**

```
Code  SEGMENT
      ASSUME CS: Code
Back:  MOV  AL, 06H
      OUT  86H, AL      ; BSR Command (Send 0) = 0000 0110
      CALL Delay
      MOV  AL, 07H
      OUT  86H, AL      ; BSR Command (Send 1) = 0000 0111
      JMP  Back
      INT  03H

Code  ENDS
      END
```

### 6.8.4 Program to Generate Rectangular Wave

**Q.** WAP to generate a rectangular wave with a 25% duty cycle on a display device connected to PortC<sub>3</sub> using BSR command.

**Soln. :**

```
Code  SEGMENT
      ASSUME CS: Code
Back:  MOV  AL, 06H
      OUT  86H, AL      ; BSR Command (Send 0) = 0000 0110
      CALL Delay
      CALL Delay
      CALL Delay
      MOV  AL, 07H
      OUT  86H, AL      ; BSR Command (Send 1) = 0000 0111
      CALL Delay
      JMP  Back
      INT  03H

Code  ENDS
      END
```

**6.8.5 Program to Generate Staircase Waveform**

**Q.** Generate a Staircase waveform on a display device connected to Port A.

**Soln. :**

```

Code    SEGMENT
ASSUME CS: Code
MOV AL, 80H
OUT 86H, AL           ; Initialize Port A as Output port
Back:   MOV AL, 00H
        OUT 80H, AL
        CALL Delay
        INC AL
        JNZ Back
        INT 03H
Code    ENDS
        END

```

**6.8.6 Program to Generate Ramp Waveform**

**Q.** Generate a Ramp waveform on a display device connected to Port A.

**Soln. :**

```

Code    SEGMENT
ASSUME CS: Code
MOV AL, 80H
OUT 86H, AL           ; Initialize Port A as Output port
Back:   MOV AL, 00H
        OUT 80H, AL
        INC AL
        JNZ Back
        INT 03H
Code    ENDS
        END

```

**Note :** Additionally the circuit will require a DAC to produce a smooth waveform like a ramp. Refer later chapters.





### 6.8.7 Program to Generate Saw-tooth Waveform

**Q.** Generate a Saw-tooth waveform on a display device connected to Port A.

**Soln. :**

```
Code    SEGMENT
ASSUME CS: Code
MOV AL, 80H
OUT 86H, AL           ; Initialize Port A as Output port
Back:   MOV AL, 00H
        OUT 80H, AL
        INC AL
        JMP Back
        INT 03H
Code    ENDS
        END
```

**Note :** Additionally the circuit will require a DAC to produce a smooth waveform like a saw-tooth.

### 6.8.8 Program to Generate Triangular Waveform

**Q.** Generate a triangular waveform on a display device connected to Port A.

**Soln. :**

```
Code    SEGMENT
ASSUME CS: Code
MOV AL, 80H
OUT 86H, AL           ; Initialize Port A as Output port
Back:   MOV AL, 00H
        OUT 80H, AL
        INC AL
        JNZ Back      ; 00H ..... FFH
        MOV AL, 0FEH
Back2:  OUT 80H, AL
        DEC AL
        JNZ Back2     ; FEH ..... 01H
        JMP Back
        INT 03H
Code    ENDS
        END
```

**Note :** Additionally the circuit will require a DAC to produce a smooth waveform like a triangular wave.



## 6.9 University Questions and Answers

Dec. 2015

Q. 3(a) Explain with block diagram working of 8255 PPI. (Refer section 6.3)

(10 Marks)

Dec. 2017

Q. 1(d) Discuss control word format for Bit Set Reset (BSR) mode of 8255 PPI. (Refer section 6.5)

(5 Marks)

May 2018

Q. 2(b) Explain PPI 8255 with block diagram. (Refer section 6.3)

(10 Marks)

Dec. 2018

Q. 4(b)(i) Explain the I/O mode control word format of 8255 PPI. (Refer section 6.4)

(5 Marks)

May 2019

Q. 3(b) Draw and explain the block diagram of 8255 Programmable Peripheral Interface (PPI) with control word formats. (Refer section 6.3)

(10 Marks)

Chapter ends...

□□□